

DEFINITION (DISCRETE CASE)

The conditional expectation of X given $Y=y$ is defined as $E[X|Y=y] = \sum x \cdot P(X=x|Y=y)$

DEFINITION (CONTINUOUS CASE)

If X, Y are conti, the conditional expectation of X given $Y=y$ is defined as $E[X|Y=y] = \int_{\mathbb{R}} x f_{X,Y}(x|y) dx$

REMARK

If X, Y are indep, then $E[X|Y=y] = E[X] \quad \forall y \in \text{Range}(Y)$

↳ Indep $\Rightarrow$ knowing Y doesn't affect the dist of X , $P_{X|Y}(x|y) = P_X(x)$, $f_{X|Y}(x|y) = f_X(x)$

EXAMPLE

Let $X, Y \sim \text{indep}$, where $X, Y \sim \text{Bin}(n, p)$

What is $E[X|X+Y=m]$?

Solution

Firstly, what is $P_{X+Y}(k|m)$?

By def, $P_{X+Y}(k|m) = P(X=k | X+Y=m) = \frac{P(X=k \cap X+Y=m)}{P(X+Y=m)} = \frac{P(X=k, Y=m-k)}{P(X+Y=m)}$

Recall, $X, Y \sim \text{Bin}(n, p)$, indep, so $X+Y \sim \text{Bin}(2n, p) \Rightarrow P(X+Y=m) = \binom{2n}{m} p^m (1-p)^{2n-m}$

Notice too, by indep, $P(X=k, Y=m-k) = P(X=k)P(Y=m-k) = \binom{n}{k} p^k (1-p)^{n-k} \binom{n}{m-k} p^{m-k} (1-p)^{n-m+k} = \binom{n}{k} \binom{n}{m-k} p^m (1-p)^{2n-m}$

$$\therefore P_{X+Y}(k|m) = \frac{\binom{n}{k} \binom{n}{m-k} p^m (1-p)^{2n-m}}{\binom{2n}{m} p^m (1-p)^{2n-m}} = \frac{\binom{n}{k} \binom{n}{m-k}}{\binom{2n}{m}}, \text{ which doesn't dep on } p$$

This is the hypergeometric distribution! Namely, $X|X+Y=m \sim \text{Hypergeometric}(2n, n, m)$ ↖ n good balls
↗ m sampled w/o replacement

$$\text{Thus, } E[X|X+Y=m] = \frac{n}{2n} \cdot m = \frac{m}{2}$$

EXAMPLE

$$\text{Let } f_{X,Y}(x,y) = \begin{cases} \frac{e^{-x/y} e^{-y}}{y}, & x, y > 0 \\ 0, & \text{o/w} \end{cases}$$

What is $E[X|Y=y]$?

Solution

To compute $f_{X|Y}(x|y) = \frac{f_{X,Y}(x,y)}{f_Y(y)}$, we first need to compute $f_Y(y) = \int_{\mathbb{R}} f_{X,Y}(x,y) dx$

$$\Rightarrow f_Y(y) = \int_{\mathbb{R}} f_{X,Y}(x,y) dx = \int_0^{\infty} \frac{e^{-x/y} e^{-y}}{y} dx = [-e^{-x/y} e^{-y}]_{x=0}^{x=\infty} = e^{-y}$$

$$\therefore f_Y(y) = \begin{cases} e^{-y}, & y > 0 \\ 0, & y \leq 0 \end{cases} \Rightarrow f_{X|Y}(x|y) = \frac{f_{X,Y}(x,y)}{f_Y(y)} = \begin{cases} \frac{e^{-x/y}}{y}, & x, y > 0 \\ 0, & \text{o/w} \end{cases}$$

That is, $X|Y=y \sim \text{Exp}(\frac{1}{y}) \Rightarrow E[X|Y=y] = \frac{1}{1/y} = y$

OBSERVE

$E[X|Y=y]$ is a function of y , $E[X|Y=\cdot] : \text{Range}(Y) \rightarrow \mathbb{R}$

$\therefore$ We can substitute the random variable Y itself to get a new rand var $E[X|Y]$

PROPOSITION

Given two rand vars X, Y , we have $E[E[X|Y]] = E[X]$ ↖ average over vals of Y

Proof (Discrete Case) ↖ average over vals of X conditioning Y

$$\text{By def, } E[X|Y] = \sum x P_{X|Y}(x|y) \text{ and } E[E[X|Y]] = \sum_y E[X|Y=y] P_Y(y) = \sum_y \left(\sum x P_{X|Y}(x|y) \right) P_Y(y) = \sum_y \sum x P_{X,Y}(x,y) = \sum_x \sum_y x P_{X,Y}(x,y) \stackrel{\text{Fubini}}{=} \sum_x x \left(\sum_y P_{X,Y}(x,y) \right) = \sum_x x P_X(x) = E[X] \quad \square$$

EXAMPLE

A shop receives a rand number N customers per day, $N \sim \text{Poi}(500)$

Each customer spends a unif rand amt of money between \$0 and \$1000, indep of the actions of all other customers.

What is the expected daily income?

Solution

Let X_i be the amt spent by the i -th customer $\Rightarrow X_i \sim \text{Unif}([0, 1000])$

Let N be the # customers $\Rightarrow N \sim \text{Poi}(500)$

Let X be the total income, $X = \sum_{i=1}^N X_i$

By Prop, $E[X] = E[E[X|N]]$

Notice, $E[X|N=n] = E[\sum_{i=1}^n X_i] = \sum_{i=1}^n E[X_i] = 500n$

$\therefore E[X] = E[500N] = 500E[N] = 500(500) = 250000$

$\uparrow$ lin of exp

EXAMPLE

We have r big tech companies. The i -th company has n_i billion dollars.

Two companies go to court, for copyright infringement, with each company equally likely to win, indep of each other

The loser pays the winner \$1 billion

This repeats until all but one company is bankrupt

What is the expected number of court cases? (If we choose the suing companies unif at rand)

Solution

$\uparrow$ in this ecosystem, all the money is only between these two companies

Case: $r=2$ (fix $n_1 + n_2 = n$)

Let $X_i := \#$ cases if company 1 starts w/ $n_i = i$

Let $Y := \begin{cases} 1, & \text{case 1 won by company 1} \\ 0, & \text{case 1 lost by company 1} \end{cases}$

Let $A_i := \#$ addnl rounds after the first if $n_i = i$

Then, $\uparrow$ a company is already bankrupt

$X_i = \begin{cases} 0, & i=0 \text{ or } n \\ 1+A_i, & 1 \leq i \leq n-1 \end{cases}$

Let $m_i = E[X_i]$, then by prop, $E[X_i] = E[E[X_i|Y]] = \frac{1}{2}E[X_i|Y=1] + \frac{1}{2}E[X_i|Y=0]$

Notice, $E[X_i|Y=1] = E[1+A_i|Y=1] = 1 + E[A_i|Y=1]$

$\uparrow$ winning case 1 by comp 1 $\Rightarrow$ basically comp 1 start with \$1 bil more, and vice versa

However, $A_i|Y=1 \sim X_{i+1} \Rightarrow E[X_i|Y=1] = 1 + E[X_{i+1}] = 1 + m_{i+1}$

$A_i|Y=0 \sim X_{i-1} \Rightarrow$ Similarly, $E[X_i|Y=0] = 1 + m_{i-1}$

$\therefore m_i = E[X_i] = \frac{1}{2}(1 + m_{i+1}) + \frac{1}{2}(1 + m_{i-1})$

$\Rightarrow m_i = \begin{cases} 1 + \frac{1}{2}m_{i-1} + \frac{1}{2}m_{i+1}, & 1 \leq i \leq n-1 \\ 0, & i=0, n \end{cases}$

$\Rightarrow m_i = i(n-i)$

General Case

Let the total # money $n = \sum_{i=1}^r n_i$

Let $X_i := \#$ court cases involving company i

$X := \#$ cases $\Rightarrow X = \frac{1}{2} \sum_{i=1}^r X_i$

$\uparrow$ every case counted twice

By lin of exp, $E[X] = \frac{1}{2} \sum_{i=1}^r E[X_i]$

Observe, X_i 's dist follows the 2 company dist since for each company, we just see "we gain money" or "we lose money to others"

$\therefore E[X_i] = n_i(n-n_i) \Rightarrow E[X] = \frac{1}{2} \sum_{i=1}^r n_i(n-n_i) = \frac{1}{2}(n^2 - \sum_{i=1}^r n_i^2)$

EXAMPLE

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Grading oral exams. (Grades are given immediately) like with online streaming algo for y'all CS nerds
The prof sees n students one-by-one, and wants to give one A^+ to the best student

Strategy: Give an A^+ to the first student $\Rightarrow P(\text{success}) = P(\text{first student is the best}) = \frac{1}{n}$ (not good)

Better strategy: See the first k students, and give no A^+ s. Give an A^+ to a student if they are better than everyone before them

Define $S = \begin{cases} 1, & \text{we give } A^+ \text{ to the best student} \\ 0, & \text{o/w} \end{cases} \Rightarrow P(\text{success}) = E[S]$

Let X = position of best student

By prop, $P(\text{success}) = E[S] = E[E[S|X]] = \sum_{i=1}^n E[S|X=i] \overset{X \sim \text{unif}}{P(X=i)} = \sum_{i=1}^n E[S|X=i] \left(\frac{1}{n}\right)$ (continue next lecture)